Migration, Childhood Parental Separation, and Adult Fertility in China：Evidence from CHARLS

Research Proposal

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Tingrui Zhang

Division of the Social Sciences, University of Chicago

1. Introduction

China's rural-to-urban labor migration boom since the 1980s and the strict access policies in most major cities created large cohorts of "left-behind children", who were left behind in their original communities during childhood as their parents worked in distant cities. The scale of this phenomenon is increasingly significant: as of 2010, the census-based estimate of rural left-behind children is approximately 61 million, and this number further increased to 67 million in 2020. The latter is roughly 38 percent of all rural children and 22 percent of all Chinese children under eighteen at that time. When these left-behind children grow up, their separation-shaped adult outcomes could significantly impact the Chinese society as a whole, as literature repeatedly found significant long-term effect of early childhood interventions (Cunha and Heckman, 2007; Heckman and Mosso, 2014). Such outcomes are starting to become visible as the first cohorts of the left-behind children are entering their 40s, and one particularly policy-relevant and academically understudied long-term effect is fertility. China's newborns fell from 16.87 million to 9.54 million between 2014 and 2024, reflecting a sharp fertility decline. Therefore, establishing the potential links between migration/childhood parental separation and adult fertility could meaningfully inform policymakers of the long-run externalities of migration policies, and inform households of the potential consequences when they make migration/separation decisions.

In light of this, this paper proposes the following research questions:

1. What is the effect of parental migration on children's fertility in China, with childhood parental separation being the key mechanism?

2. Through what channel does this effect operate? Is it due to the separation's influence on human capital, or its influence on intergenerational support?

In an attempt to identify migration's influence and the above two channels, this paper builds a household model involving three generations. The first generation (i.e., G0) decides the second generation (i.e., G1)'s left-behind status through a utility maximization process influenced by a number of covariates, which shifts the human capital and family attachment of G1. With these inherited, G1 then decides fertility (number of G2) through a utility maximization problem similar to that of G0, generating the outcome of interest in this paper.

The China Health and Retirement Longitudinal Study (CHARLS) is used to study this three-generation chain because each respondent in the 45+ panel reports on every adult child individually, including whether each child was separated from their parents. It also contains detailed family characteristics, including income, intergenerational transfers, and family attachment variables, allowing decomposition of the left-behind effect on fertility.

The empirical strategy combines two complementary identification designs that probe causality from different angles. The primary design uses an instrumental variable to identify the effect of a migration-cost shock on the adult fertility of exposed rural-origin cohorts. Specifically, this paper uses county-level rollout-year variation in national identity-card distribution, which differentially lowered the legal cost of temporary urban residence for rural migrants (de Brauw and Giles, 2017). ID-rollout is a documented exogenous shock with clean policy implications, which affects G1's fertility through increased migration probability and thus increased separation probability. This paper also employs a second IV as a robustness check: a county-by-destination Bartik shift-share design (Goldsmith-Pinkham, Sorkin, and Swift 2020), using pre-existing migration shares interacted with destination-city migrant-children school-access policy. Both instruments identify migration-cost-shock or migration-policy-induced family-arrangement bundle effects, with the reduced-form regression of fertility on the primary instrument as the headline within-design output. The second design proxies the causal effect of childhood parental separation by exploiting within-family fixed effects, using sibling differential exposure to migration/separation across pre-school and school-age windows, and controlling for family-level selection into migration. With within-family controls mitigating endogenous family choice of parental separation, this design identifies a within-family conditional contrast across siblings, and tests for different channels of the effect.

The contribution is, to my knowledge, one of the first three-generation analyses linking migration/childhood parental-separation exposure to adult fertility and intergenerational exchange in China, building on three antecedent literatures that have not yet been bridged. First, the China left-behind empirical literature (Hu 2013; Zhou, Murphy, and Tao 2014; Wang 2014; Lu 2014; Meng and Yamauchi 2017) establishes that childhood left-behind status reduces cognitive achievement, worsens non-cognitive outcomes, and delays first birth age. These studies mostly focus on human capital outcomes, with no completed fertility results observed. Second, the human capital formation literature (Cunha and Heckman 2007; Cunha, Heckman, and Schennach 2010) concludes that early investment in human capital makes later investment more effective, but has not applied the framework to the left-behind children scenario. Third, the structural fertility literature (Schoonbroodt and Tertilt 2014; De La Croix and Doepke 2003; Vogl 2016; Kim, Tertilt, and Yum 2024) found meaningful impacts of cultural phenomena like filial piety on fertility choice in equilibrium, but did not consider the effect of migration/childhood parental separation.

The proposal proceeds as follows. Section 2 reviews the relevant literature, organized around the model's motivations. Section 3 develops the three-generation household model. Section 4 describes the CHARLS data, and discusses the construction of treatment variables and sample feasibility. Section 5 presents the reduced-form identification strategy. Section 6 concludes. References follow.

2. Literature review

The literature review is organized around the motivations of the household model in Section 3. Each subsection of the review anchors a strand of literature to a specific theoretical object that the model needs to address or incorporate.

2.1. The China left-behind empirical literature

2.1.1. Consequences of childhood parental separation

The China left-behind literature provides priors on the sign of the left-behind coefficient on child human-capital outcomes. Hu (2013) studies left-behind children's schooling in rural northwest China using a household-level matching design. Zhou, Murphy, and Tao (2014) document that left-behind children in rural Anhui have lower test scores than non-left-behind peers, with the gap concentrated among children separated in early childhood. Wang (2014) uses historical village migrant networks as an instrument for parental migration and finds a negative effect on schooling years. Meng and Yamauchi (2017) document negative cumulative effects of parental migration on child education and health using the 2009 Rural-Urban Migration in China survey. Zhang, Behrman, Fan, Wei, and Zhang (2014) found that being left-behind by both parents reduces children's contemporary achievements by 5.4 percentile points for math and 5.1 percentile points for Chinese. Wen and Lin (2012) and Zhao, Yu, Wang, and Glauben (2014) document elevated depression, anxiety, and externalizing-behavior scores in left-behind children. The literature converges on a negative direction of short-term effect on cognitive and non-cognitive outcomes, with effects graded by age at and duration of exposure. This is incorporated in this paper's model as the negative human capital effect of childhood parental separation.

The literature on adult outcomes of left-behind exposure is thinner. Antman (2012) reviews the available evidence and concludes that adult outcomes are largely unstudied. Chang, Dong, and MacPhail (2011) document persistent labor-market gaps for adults reporting childhood left-behind exposure. The closest published comparison is Zhao, Lu, and Zhang (2025), who used the China Family Panel Study (CFPS) to study the effect of left-behind experience on human capital accumulation and age of first birth. Using an instrumental variable design, they find that left-behind experiences delay the age of first birth by reducing individuals' physical health levels and increasing their openness personality traits in adulthood. However, the channels of left-behind experience's impact on first birth age and complete fertility are very different. For instance, the intergenerational support channel proposed in this paper is not considered in Zhao, Lu, and Zhang (2025). Additionally, the empirical effect in this paper is backed by theoretical evidence, which is also a major contribution that is not seen in previous work.

2.1.2. Grandparental childcare in China

The family-attachment channel in Stage 2 is motivated by the central role of grandparents in childcare provision. When adult children face work or migration constraints, grandparents often substitute for parental childcare and thereby reduce the effective cost of child-rearing. Prior evidence supports this mechanism: Compton and Pollak (2014) show that grandmother proximity is associated with adult-daughter labor supply in the United States, while Du, Dong, and Zhang (2019) document that grandparental childcare is the dominant childcare arrangement in urban China and is positively associated with adult daughters' labor-force participation. Lei, Strauss, Tian, and Zhao (2015) further document widespread intergenerational coresidence in rural China using CHARLS. At the same time, grandparental care is not a perfect substitute for parental care. Zhang et al. (2021), using CFPS, find that children cared for by grandparents experience delays in early childhood development relative to those receiving parental care. This evidence suggests that grandparents often fill the childcare gap created by parental migration or parental overwork, but that the quality and availability of this substitution depend on the strength of intergenerational family attachment. The model therefore allows grandparental support to lower the effective cost of child-rearing, with the magnitude of this support determined by family attachment.

2.2. The human capital formation literature

The human-capital production function of Section 3 places two parameters on the left-behind indicator: the direct early-childhood human-capital loss from separation, and the modifier of the return to investment in later stages. The discipline behind this specification comes from Cunha and Heckman (2007) and Cunha, Heckman, and Schennach (2010), who introduce the technology of skill formation in which adult human capital is produced by recursive application of a CES production function over multiple childhood stages. The framework has two empirically central features: self-productivity (skill at stage k raises skill at stage k+1) and dynamic complementarity (the return to investment at stage k+1 is increasing in skill produced at stage k). Heckman (2007) and Heckman and Mosso (2014) extend the framework to non-cognitive skills and emphasize that the return to investment is age-graded, with earlier-stage investment producing larger returns than later-stage investment. Therefore, this paper models childhood parental separation not only as harmful to human capital levels but also to the efficiency of later investment.

2.3. The structural fertility literature

The fertility choice problem of this paper's model builds on De La Croix and Doepke (2003), who introduce a quantity–quality trade-off in which poor parents choose higher fertility and lower per-child education, and on Vogl (2016), who extends the framework to a multi-generation setting and shows that quality–quantity selection alone can generate persistent human-capital inequality across generations. The most recent quasi-experimental test is Rossi and Godard (2022), who exploit the 1990s extension of social pensions in Namibia and find that pension receipt substantially reduces fertility, especially in late reproductive life. Schoonbroodt and Tertilt (2014) argued that fertility responds to rules governing the direction and strength of intergenerational transfers, not only to wages and child-rearing costs. When parents partially own property rights over their children, they expect transfers from their children. When parents have weaker enforcement capacity, their fertility falls because children become a less effective old-age investment. This is incorporated in the model of this paper by allowing family attachment to affect the intergenerational transfer bounds. Notably, none of the previous fertility models considered the role of childhood separation, which is a novelty of this paper.

3. A three-generation household model

The model has two stages and three generations. G0 is the CHARLS-respondent generation (aged 45+ at survey). G1 is the adult-child generation, some of whom were exposed to parental migration/separation in the pre-school and/or school-age windows (the empirical proxy for childhood parental separation in this paper). G2 is the grandchild generation, born to G1. Stage 1 is G0's choice over consumption, fertility, child human-capital investment, intergenerational transfers, and the left-behind status. Stage 2 is G1's choice over consumption, fertility (the count of G2), and child human-capital investment, taking as given the human capital and family attachment inherited from Stage 1.

3.1. Notation

The following notation is used throughout. Each symbol is indexed by stage where applicable; cross-stage parameter equality is stated explicitly when assumed.

Variable	Meaning and Construction
c_0, c_1	G0 consumption (Stage 1) and G1 consumption (Stage 2)
n_0, n_1	G0 fertility (number of G1 children) and G1 fertility (number of G2 grandchildren)
e_0, e_1	Per-child human-capital investment in Stage 1 and Stage 2
h_0, h_1, h_2	Human capital of G0, G1, G2 respectively
k	Per-child intergenerational transfer in Stage 1 (G0 to G1 if positive; G1 to G0 if negative)
LB	Binary parental-separation indicator for G1, 1 if parents were not the primary caregiver in either the pre-school or school-age window, 0 otherwise. Constructed from CHARLS Life History 2014 c034 caregiver indicators.
a_1	Family attachment between G1 and G0. More family attachment reduces G1's childcare costs, through G0 grandchild care for G2. Empirical counterpart in CHARLS combines CF001-003 (G0 grandchild care, genuinely G0 → G2) with CA018 (child-lineage downward financial support).

3.2. Stage 1: G0's separation, fertility, and investment choice

G0 enters Stage 1 with human capital h_0 and chooses consumption c_0, fertility n_0, per-child investment e_0, per-child transfer k_1, and a binary migration / left-behind status LB ∈ {0,1} for each child. The basic cost per child is f, and an additional e_0 is required for human capital investment per child. Raising one child consumes a proportion τ of parental time. Standardizing parental time to 1 unit, parental working time is $1 - \tau n_0$. Parents possess h_0 units of human capital. G0's labor income is shifted by the migration wage premium b when LB = 1, which does not represent a literal change in G0's stock of human capital. Standardizing the wage per unit of human capital to 1, G0's labor income is

$$(1 - \tau n_0)(1 + bLB)h_0$$

Parents can make transfer payments to each child in the amount of k_1. If parents have a positive net transfer to their children, then $k_1 \geq 0$. Conversely, transfer payments from children to parents due to filial piety are termed "upward transfers," corresponding to the case where $k_1 < 0$.

Therefore, the budget constraint of G0 can be expressed as:

$$c_0 + (f + e_0 + k_1)n_0 = (1 - \tau n_0)(1 + bLB)h_0$$

To bound transfers and capture the effect of childhood separation on transfers, this paper further introduces a family attachment variable $a1$, which decreases if G0 decides to migrate and leave G1 behind. Empirically, a_1 is operationalized as a composite of two CHARLS items, reported separately in the mechanism analysis: G0 grandchild care for G2 (CF001-003 in earlier-wave terminology, hours of care per year) and G0 downward financial support flowing toward each non-co-resident child's lineage (CA018 in CHARLS wave 5).

$$a_1 = a_0 + \theta_a LB$$

This paper draws upon Schoonbroodt and Tertilt (2014)'s framework where parents partially own property rights over their children, allowing the intensity of authoritative filial piety to determine the lower bound of negative transfers. Additionally, this proposal lets a_1 affect the lower bound of such transfer, with lower a_1 indicating a tighter lower bound. So if a child is left behind, future family attachment is weaker, and thus G0 cannot expect or enforce as much future upward transfer from G1:

$$k_t \geq -x(a_1)$$

with

$$x'(a_1) > 0.$$

Per-child human-capital production for G1 is given by

$$h_1 = (\theta_0 + \theta_1 e_0 + \theta_2 LB + \theta_3 LBe_0)^{\delta} h_0^{\gamma}.$$

Here, h_1 represents the human capital level of the child. The parameter θ_0 denotes the child's human capital endowment, while θ_1 signifies the efficiency of human capital investment. $\theta_2 < 0$ captures the direct early-childhood human-capital loss from separation, and $\theta_3 < 0$ captures "inverse dynamic complementarity" in the Cunha-Heckman sense: when G1 is left behind, the return to subsequent investment e_0 is lower because the early-stage human-capital stock that complements later-stage investment has been damaged. The bracket is restricted to the region where $\theta_0 + \theta_1 e_0 + \theta_2 LB + \theta_3 LBe_0 > 0$; the empirical implementation enforces this restriction via lower-bound parameter constraints on (θ_0, θ_1) such that the bracket remains positive at LB = 1 for plausible e_0 ranges, and a robustness specification replacing the production function with the exponential form $h_1 = \exp(\theta_0 + \theta_1 e_0 + \theta_2 LB + \theta_3 LBe_0)^{\delta} h_0^{\gamma}$ will be used if necessary.

The total income of G1, s_1, comprises two components: the offspring's human capital h_1 and parental transfers to offspring k_1:

$$s_1 = h_1 + k_1$$

Putting everything together, G0 chooses their consumption c_0, number of children n_0, children's transfers k_1, investment in children e_0, and separation/migration status LB to maximize their utility U_0, composed of c_0, n_0, s_1, and family attachment a_1:

$$\max_{c_0,n_0,e_0,k_1,LB} \; U_0 = \ln c_0 + \alpha(\ln n_0 + \beta \ln s_1) + \mu \ln (a_1)$$

subject to:

$$s_1 = h_1 + k_1$$
$$a_1 = a_0 + \theta_a LB$$

$$h_1 = (\theta_0 + \theta_1 e_0 + \theta_2 LB + \theta_3 LB e_0)^{\delta} h_0^{\gamma}$$
$$c_0 + (f + e_0 + k_1)n_0 = (1 - \tau n_0)(1 + bLB)h_0$$
$$k_1 \geq -x(a_1), x'(a_1) > 0$$

The binary participation choice LB ∈ {0,1} is governed by the discrete value comparison

$$LB = 1 \Leftrightarrow U_0(LB = 1) - U_0(LB = 0) > 0$$

3.3. Stage 2: G1's fertility and investment choice

G1 enters Stage 2 with three state variables determined or shifted by Stage 1: human capital h_1, given by the stage-1 production function; family attachment variable a_1, which is affected by migration/separation and further influences resources flowing from G0 to G1 (and to G2) that reduce the cost of raising G2; and transfer k_1, which could either be the negative or positive: if $k_1 > 0$, it increases G1's resources through downward transfers from G0; if $k_1 < 0$, it represents an upward filial-transfer obligation and reduces G1's disposable resources. Adult G1 chooses consumption c1, number of children n1, and child-quality investment per child e1 to maximize their utility:

$$U_1 = \ln\, c_1 + \phi \ln (1 + n_1) + \psi \cdot I(n_1 \geq 1) \cdot \ln\, h_2$$

with $\ln (1 + n_1)$ replacing the standard ln n_1 to accommodate the zero-fertility corner solution $n_1 = 0$ (childless adults, an outcome of substantive empirical interest given Chinese fertility decline) and with $\psi \cdot I(n_1 \geq 1) \cdot \ln\, h_2$ including an indicator for having at least one G2 child to handle the undefined h_2 when no G2 exists. In the baseline model, ϕ and ψ are treated as constant, so LB affects G1 fertility only through inherited human capital, family attachment, and transfers. A richer extension could allow ϕ and ψ to depend on LB to capture an additional preference channel, as literature suggests that childhood adversity directly shifts adult preferences over family formation (Heckman, Pinto, and Savelyev 2013; Conti, Heckman, and Pinto 2016).

The Stage 2 budget constraint is:

$$c_1 + n_1(f + e_1 - \omega a_1) = (1 - \tau n_1)h_1 + k_1$$

Which includes ωa_1, the per-G2-child cost reduction provided by G0 support inflow, and k_1, the transfer between G0 and G1. G2's human-capital production is

$$h_2 = (\alpha_0 + \alpha_1 e_1)^{\delta} h_1^{\gamma}$$

The cross-generation parameters δ and γ are assumed equal to their Stage 1 values, although a sensitivity analysis allows them to vary by stage. Overall, G1's maximization problem is:

$$\max_{c_1,n_1,e_1} U_1 = \ln\ c_1 + \phi \ln\left(1 + n_1\right) + \psi \cdot I(n_1 \geq 1) \cdot \ln\ h_2$$

subject to:

$$c_1 + n_1(f + e_1 - \omega a_1) = (1 - \tau n_1)h_1 + k_1$$

$$h_2 = (\alpha_0 + \alpha_1 e_1)^{\delta} h_1^{\gamma}$$

The baseline model does not involve the LB term in Stage 2. This simplification is made because the paper's main outcome is G1's fertility, not G2's living arrangement. Instead, the model captures this margin through the effective cost of child-rearing: when G1 faces work or migration constraints, support from G0 can substitute for parental childcare and lower the cost of having children. The availability of this support depends on family attachment. Thus, the location and childcare arrangement of G2 is absorbed into the grandparental-support channel rather than modeled as a separate decision. Another reason is that there is no second-generation left-behind treatment in the data, so adding LB in stage 2 would make the model infeasible to calibrate. Lastly, as many G1 migrate to live with their parents in cities eventually, migration/separation is less prevalent in the G1 generation. This abstraction is thus empirically reasonable as a baseline, while future work could build richer models allowing G1 to choose whether to migrate and bring G2 with them.

3.4. Channels of migration/separation's impact on fertility

The model produces two testable channels through which LB affects G1 fertility. First, the human-capital channel: LB reduces h_1 through $\theta_2 < 0$ and reduces the return to e_0 via $\theta_3 < 0$; lower h_1 reduces both the income available for raising G2 and the opportunity cost of childbearing, so the net sign on n_1 is theoretically ambiguous. Second, the family attachment channel: LB reduces a_1 through $\theta_a < 0$. Lower a_1 raises the per-G2-child cost (because ωa_1 is smaller), which reduces n_1. However, lower a_1 tightens the lower bound on k_1, making k_1 less negative. This may raise n_1 because it increases G1's disposable resources. Therefore, the family attachment channel's effect on fertility is also ambiguous. Additionally, a preference channel may exist if $\phi'(LB) \neq 0$ or $\psi'(LB) \neq 0$: if childhood adversity directly reduces the adult preference for fertility, this further reduces n_1. While testing $\phi'(LB) = \psi'(LB) = 0$ empirically is not feasible given current data, it is testable if this model is fully calibrated. The aggregate reduced-form coefficient β on LB in the fertility equation is the sum of these channels.

4. Data and Sample

This study uses the China Health and Retirement Longitudinal Study (CHARLS), a nationally representative longitudinal survey of middle-aged and older adults in China. CHARLS is particularly suitable for this project because it links information across three generations: the respondent generation

(G0), their adult children (G1), and their grandchildren (G2). The core survey provides detailed information on respondents' demographic characteristics, hukou status (i.e., residential place status), household income, health, intergenerational transfers, living arrangements, and family support. This section explores the data and discusses how key variables are constructed and validated.

4.1. The treatment variable

The migration/separation treatment is constructed from the CHARLS Life History 2014 module's c034 series and the migration/hokou series. c034 records, separately for each of up to eighteen biological children of the respondent, who took primary care of the child during pre-school years (item c034c1) and during school-age years (item c034d1). Each c034 item is a multiple-response array with caregiver categories s1 through s10: s1 biological mother, s2 biological father, s3 foster or stepmother, s4 foster or stepfather, s5 grandparents (paternal side), s6 maternal grandparents, s7 siblings, s8 other relatives, s9 nanny, s10 others. Treatment = 1 requires that child i in family j is separated from parents during a given age window (i.e., grandparents, siblings, other relatives, nanny, or others (s5, s6, s7, s8, s9, s10) are recorded as a primary caregiver during that window). An important fact is that the c034 separation indicator captures any case where parents were not primary caregivers of the child during pre-school or school-age windows, regardless of the cause. Cases of c034 non-parent care include (i) parental migration for work, the canonical left-behind case; (ii) maternal labor-force participation with no migration but with grandparents providing day care while parents worked; (iii) parental death or divorce and grandparents assuming primary care; (iv) traditional intergenerational coresidence in which grandparents were primary day-time caregivers but parents were geographically present; and (v) other non-migration arrangements.

To address this issue, two additional requirements are added for treatment to be equal to 1. (i) The analytic sample is restricted to G0 with rural hukou (CHARLS 2013 bc001 = 1, agricultural hukou), which removes most non-migration sources of non-parent care: urban G0 with grandparent-care arrangements are most likely traditional intergenerational coresidence cases, while rural G0 with non-parent care arrangements are most likely migration-driven. (ii) The CHARLS Life History 2014 Work_History module records G0 employment history, including self-reported migration spells with year-by-year start and end dates. The requirement is that for each G1 child i, G0 was working in a migrant destination during the c034 non-parent care window. Where Life History or other CHARLS items record household earnings or income sources during the relevant childhood period, the presence of migrant-worker income or remittance is also reported as a robustness check.

The migration/separation treatment is operationalized as two distinct binary variables:

$$LB_{ij}^{pre} = 1$$

if the child experienced non-parent-care in the pre-school window, had rural hukou, and had parents working in migration destinations during that window;

$$LB_{ij}^{sch} = 1$$

if the child experienced non-parent-care in the school-age window, had rural hukou, and had parents working in migration destinations during that window. This decomposes the treatment into three mutually exclusive categories: pre-school-only separation, school-age-only separation, and separation in both windows. Compared with published Chinese rural left-behind literature (Hu 2013; Wang 2014; Lu 2014), which has typically operationalized LB as a binary contemporaneous indicator, this design allows the separation effect to vary across two windows of childhood, capturing richer dynamics.

4.2. Cohort feasibility and sample size

An unavoidable design tension is that not all G1 cohorts in the survey are exposed to the migration boom from the 1980s to 2000s. The analytic design must specify which cohorts carry identifying weight for which outcome, rather than treating CHARLS G1 as a homogeneous panel. The following table reports the cohort-feasibility crosstab:

Table 1

Cohort Feasibility

G1 birth cohort	Age in 2018	Modern LB exposure	Fertility outcome usable?	Role in design
1955–1964	54–63	Predates modern migration	Completed fertility	Mostly excluded; non-migration separation construct
1965–1974	44–53	Partial; pre-1990s	Completed fertility	Secondary cohort; pre-modern LB
1975–1984	**34–43**	**Yes; childhood overlaps 1990s migration**	**Women 40+ near-completed; men 34–43 still fertile**	**Candidate core cohort for women's near-completed fertility (40+) and adult observed parity, subject to realized sample size; male outcomes interpreted as observed parity, not completed fertility**
1985–1994	24–33	Yes; modern LB	Fertility censored; timing only	Used for fertility-timing outcomes (birth by 25, by 30, ever-parent)

The candidate core identifying cohort is G1 born 1975–1984, who were children during the 1980s to 1990s migration boom and whose adult fertility outcomes can be observed by survey age, with the sex-specific qualification that women 40+ at survey are treated as having near-completed fertility, while men 34–43 are still partly within their reproductive window and reported as observed parity rather than completed fertility.

The CHARLS analytic sample begins from approximately 17,500 G0 respondents in CHARLS wave 1 (2011), with subsequent waves adding refresher respondents in 2013, 2015, and 2018. The Life History 2014 supplement (which contains the c034 treatment items) was administered to 20,543 respondents, of whom 16,227 also appear in CHARLS 2018. The c034 series is reported by 11,142 of 20,543 Life History 2014 respondents, generating 32,370 child-records with caregiver data. After merging with CHARLS 2018 Family_Information.dta for child birth-year and gender (xchildbirth, xchildgender) and CHARLS 2013 Demographic_Background.dta for G0 hukou (bc001), the analytic dataset contains 16,595 child-records with treatment, demographics, and G0 hukou jointly observed. In the rural-G0 1975-1984-cohort candidate core, childhood separation prevalence is 22.9% on a sample of 4,361 child-records; in the headline rural-women-40+ analytic sample, the prevalence is 19.1% on 3,349 child-records with non-missing fertility outcome. Both sample sizes are substantively meaningful; the prevalence is not trivially small.

4.3. One-Child Policy (OCP) Exposure

G1 fertility decisions span 1990 to 2025 across CHARLS cohorts, fully overlapping the One-Child Policy era (1979–2015) and the early years of the two-child policy (2016–2021) and three-child policy (2021–). OCP enforcement intensity varied by hukou (urban-strict, rural-flexible), by ethnicity (minority exemptions), and by provincial implementation. The baseline empirical specification controls for OCP exposure through province × hukou × G1-birth-cohort fixed effects. These fixed effects absorb province- and hukou-specific cohort differences in family-planning exposure, local fertility norms, and other cohort-specific institutional changes, without requiring an explicit functional form for OCP intensity. The saturated province × hukou × cohort fixed effects are the baseline for the family fixed effects design, where they absorb cohort-and-OCP variation while preserving within-family variation in LB. For the Bartik shift-share design, the instrument varies at the village × cohort level, and for the ID rollout design, the instrument varies at the county × cohort level. For them, province × cohort fixed effects do not mechanically absorb the instrument as long as there is within-province variation in county rollout timing. However, they remove all province-level cohort shocks and leave identification only from within-province differences in villages' baseline exposure shares. This may substantially weaken the first stage, and specifications will be reported both with and without the province × cohort component of the OCP FE to document the trade-off transparently. For the 1985–1994 cohort whose fertility decisions partially overlap the post-OCP relaxation period, the analysis treats outcomes as fertility timing rather than completed fertility (birth by age 25 / 30, ever-parent) to accommodate the censoring.

5. Empirical strategy: reduced form

The reduced-form analysis combines two complementary identification designs. The first set of designs (Design A) uses instrumental variables to exploit the causal effect of migration, which includes the ID instrument (Design A - i) that exploits migration cost reduction, and the Bartik destination-access design (Design A - ii). The second within-family fixed-effects estimator (Design B) exploits sibling differential exposure to non-parent caregiving across pre-school and school-age windows. Designs A and B identify different objects and are reported in parallel rather than as substitutes, which are discussed in detail in this section.

5.1. Design A - i: ID-rollout

5.1.1. Design and Identification

The primary instrumental-variables design uses the staggered county-level rollout of national identity cards to rural residents during 1984-1997, which is first proposed by de Brauw and Giles (2017) as a migration cost reduction policy. Following the 1988 reform of the residential registration system, possession of a national ID card became necessary for rural migrants to obtain legal temporary residence permits (zanzu zheng) in cities. A valid ID also makes migrant workers more employable as they can prove their identity without relying on hukou (village) documents. Therefore, people in places with earlier ID rollout are more likely to migrate and leave their children behind. The design treats ID-rollout as the primary identifying variation because it cleanly identifies the bundle effect of migration on children's fertility, although the effect of different channels cannot be cleanly separated. Additionally, it has strong policy implications as a clean shock to migration cost with documented exogeneity properties. The instrument is constructed as a polynomial in months-since-ID-rollout in the relevant childhood-window cohort year:

$$Z_{ivt}^{ID} = f(months\_since\_rollout_{c(v),t})$$

where c(v) is the county containing village v, $\mathrm{months\_since\_rollout}_{c(v),t}$ enters as a quartic following de Brauw and Giles (2017) Table 3 first-stage specification. The headline first stage is estimated as:

$$LB_{iv} = \pi_1 Z_{ivt}^{ID} + X_{iv}'\pi_2 + \mu_{county} + \lambda_{cohort} + u_{iv}$$

Where $LB_{iv} = 1[LB_{ij}^{pre} + LB_{ij}^{sch} > 0]$ to avoid under-identification. μ_{county} is the county-of-origin fixed effect (the identifying variation lives at the county-cohort level since rollout-year is a county attribute) and $\lambda_{\mathrm{cohort}}$ is the G1 birth-cohort fixed effect. The first-stage F is reported with weak-IV-robust adjustments. The second stage is

$$Y_{iv} = \beta \widehat{LB}_{iv} + X_{iv}'\gamma + \mu_{county} + \lambda_{cohort} + \varepsilon_{iv}$$

where Y is the cohort-and-sex-appropriate fertility outcome, and $\mathrm{L\hat{B}}$ is the first-stage prediction in the headline specification. Province × cohort fixed effects are not used as the baseline saturation because they may substantially reduce first-stage power and because the baseline specification already includes county and cohort fixed effects. However, because the ID-rollout instrument has county-level rollout-year variation, province × cohort fixed effects are not claimed to mechanically absorb the instruments. Their effect on first-stage strength will be reported once the actual instruments are constructed. The reduced form is:

$$Y_{iv} = \epsilon Z_{ivt}^{ID} + X_{iv}'\gamma + \mu_{county} + \lambda_{cohort} + \varepsilon_{iv}$$

The interpretation of the coefficients is calibrated to what the ID rollout actually shifts. The reduced-form regression coefficient ϵ is the cleanest object: it is the effect of the migration-cost shock on adult fertility outcomes, not contingent on any specific causal channel running. The reduced form is therefore reported as the primary Design A-i output, alongside the 2SLS coefficient. The 2SLS coefficient β is interpreted as the change in G1 fertility associated with an ID-rollout-induced increase in measured childhood separation. This may not be a clean local average treatment effect of separation alone because the exclusion restriction may not hold, which will be discussed in the next section. Within this framing, the policy implication is clean: ID-rollout-induced reductions in migration cost have a measurable reduced-form effect on adult fertility, regardless of whether the precise mechanism runs through parental separation or through the wider bundle of migration-related family-arrangement margins.

5.1.2. Assumptions and Robustness Checks

This IV design involves two assumptions. Firstly, the relevance assumption requires that variation in ID rollout actually changed the probability that parents migrated for work during the child's childhood. This is intuitive as ID rollout decreases migration barrier for the rural population, and has already been proven by de Brauw and Giles (2017), who documented a very strong first stage. Secondly, the exclusion restriction assumes that ID rollout does not directly affect the child's later fertility except through its effect on childhood parental migration or left-behind status. This is a stronger assumption, but it is supported by the institutional fact that the ID card primarily reduced documentation and monitoring frictions associated with temporary work outside one's hukou location. Despite this, ID rollout could also shift other factors, like the family's long-run urban settlement, that could affect fertility. To address alternative channels, I control flexibly for birth-cohort effects, origin-location fixed effects, whether the child migrates eventually, and destination-specific trends if the child migrates. The identifying assumption is that, after these controls, the remaining variation in ID rollout affects adult fertility only through childhood parental migration/separation. Notably, even if the exclusion restriction is violated, the headline reduced form remains informative about the total effect of ID-rollout exposure on later fertility.

Another concern is whether the ID rollout policy is exogenous, as the effect could be noisy if ID rollout is correlated with other policies or incentives that affect later fertility. Fortunately, this concern is largely addressed by current evidence. The Ministry of Civil Affairs, which administered ID distribution, was institutionally separate from the Ministries of Agriculture and Finance that set policies affecting land, credit, and grain procurement. Additionally, de Brauw and Giles (2017) document that timing of ID rollout across the eight provinces in their sample (Anhui, Jilin, Jiangsu, Henan, Hunan, Shanxi, Sichuan, Zhejiang) is uncorrelated with rainfall shocks affecting migration incentives, with weighted village tax rates proxying for administrative capacity, with grain quota policy implementation, with land tenure security indicators, and with collective asset management. This supports the conditional exogeneity of rollout timing once village fixed effects absorb time-invariant village characteristics.

To further test the validity of this IV, I will implement a Visual Instrumental Variable (VIV) diagnostic. Specifically, I will split the data into groups by county, perform 2SLS for each group, and plot group-specific reduced-form effects of ID-rollout exposure on adult fertility against corresponding first-stage effects on childhood parental migration. Under the exclusion restriction and a stable treatment-effect interpretation, the line should have a slope corresponding to the second stage estimate, as groups with stronger first stages should exhibit proportionally stronger reduced-form effects that are captured by the IV estimate. The line should also pass through the origin if the exclusion restriction holds, because if the instrument has no effect on treatment, it should have no reduced-form effect on the outcome (Angrist, 2022).

5.1.3. Data

The de Brauw-Giles county-level rollout-year dataset is not currently publicly archived. The Journal of Human Resources 2017 article does not have a mandated AEA-style data archive, and the World Bank Economic Review 2018 supplementary online appendix does not contain a county-by-county table. The published papers report aggregate distributions (41 of 88 counties received IDs in 1988; range 1984-1997 across the eight provinces) but not the county-level coding. The dataset must therefore be obtained by direct correspondence with the authors (Alan de Brauw at IFPRI; John Giles at the World Bank).

5.2. Design A - ii: Bartik destination-access sensitivity instrument

The Bartik destination-access design is reported as a sensitivity to Design A-i and as the alternative headline if ID-rollout data access is not achievable within the project timeline. The instrument is constructed at the village × destination × cohort level as

$$Z_{ivt} = \sum_{d} s_{v,d}^{pre} \times AccessPolicy_{d,t}$$

where v is the village of origin, d indexes destination cities, t is the relevant childhood-window cohort year, $s_{v,d}^{pre}$ is the pre-existing share of migrants from village v going to destination d (measured from the 1990 census long-form, predating the LB-relevant period for most CHARLS G1 cohorts), and $AccessPolicy_{d,t}$ is an indicator for whether destination d in year t allowed migrant children to enroll in public schools, take exams locally, or obtain urban public-service access. The Bartik design is conceptually distinct from the ID-rollout instrument, as Bartik shifts a destination-policy-induced family-arrangement bundle while ID-rollout shifts a migration-cost shock. When both instruments are available, an over-identified 2SLS is reported with a Hansen J test of over-identifying restrictions to assess the joint exclusion-restriction plausibility.

The Bartik instrument's feasibility is also constrained by external data construction needs. The 1990 census long-form county-by-destination migration shares are expected to be obtainable from NBS data channels at the county level (sufficient for a Tier 2 county × destination Bartik, conditional on access verification); village-level shares (Tier 1) are partially anonymized in the public release. The destination-city migrant-children school-access policy panel is the binding feasibility constraint: it is not contained in any single public dataset and must be constructed by manual digitization from provincial education-bureau records and the Chinese-policy-history literature, an undertaking that likely requires several months of dedicated work.

Like the ID rollout instrument, AccessPolicy shifts a bundle of family-arrangement margins through migration, including whether the child is left behind, whether the family expects long-run urban settlement, the child's hukou and schooling trajectory, and later urban employment and marriage-market exposure, of which separation is one component. The Bartik 2SLS is therefore interpreted as a destination-policy-bundle effect, parallel to but distinct from the ID-rollout migration-cost-shock effect. Agreement between the two independent identifying strategies on the reduced-form direction strengthens the inference that migration-induced childhood family-arrangement changes are associated with later fertility outcomes.

5.3. Design B: within-family fixed effects

5.3.1. Design and Identification

To address Design A's drawbacks in that it only identifies the migration bundle effect and cannot test migration/separation's effect during different windows, this proposal employs a within-family fixed effect design. This strategy exploits sibling differential exposure to LB within the same family. The estimating equation for G1 fertility is

$$Y_{ij} = \beta_1 LB_{ij}^{pre} + \beta_2 LB_{ij}^{sch} + X_{ij}'\gamma + \lambda_j + \varepsilon_{ij}$$

where LB_{ij}^{pre} and LB_{ij}^{sch} are the separation/left behind status of child i within family j during the pre-school period and school-age period, Y_{ij} is the relevant cohort-appropriate fertility outcome, λ_j is a family fixed effect, X_{ij}' is a vector of within-family-varying controls (child gender, birth order, age at survey, OCP exposure absorbed via province × hukou × G1-birth-cohort fixed effects), and ε_{ij} is the residual. The coefficients β_1 and β_2 identify within-family conditional contrast across siblings differentially exposed to non-parent caregiving during the two windows, controlling for all family-level time-invariant characteristics, including G0 unobserved migration propensity, village of origin, and family economic conditions in the LB-relevant period. This specification allows testing for whether the timing of separation matters, which is an important improvement from the literature.

The maintained assumption for causal interpretation of β_1 and β_2 is that within-family LB exposure is conditionally random across siblings given the controls. This assumption is vulnerable to three well-documented within-family confounders: birth-order effects (Black, Devereux, and Salvanes 2005), gender-of-child effects (Hu 2013 documents that sons and daughters in rural Chinese households are differentially left behind), and within-family resource reallocation in response to one sibling's separation. The proposed solution is to (i) include explicit controls for birth order, gender, and family size; (ii) report estimates on the subsample of same-gender sibling pairs, which absorbs the gender confounder; (iii) frame β_1 and β_2 as within-family conditional contrasts across siblings differentially exposed across c034 caregiver windows, conditional on observable controls, rather than as a clean within-family ATT.

In an attempt to decompose the effect of migration/separation on fertility, mechanism analysis under Design B reports separate within-family regressions of LB on G1 human capital h_1 (proxied by years of schooling) and on the G0 support inflow a_1 (G0 grandchild care for G2 from CF001-003, and child-lineage downward financial support from CA018):

$$h_{1ij} = \alpha_{h1} LB_{ij}^{pre} + \alpha_{h2} LB_{ij}^{sch} + X'_{ij}\gamma + \lambda_j + u_{ij}$$

$$a_{1ij} = \alpha_{a1} LB_{ij}^{pre} + \alpha_{a2} LB_{ij}^{sch} + X'_{ij}\gamma + \lambda_j + v_{ij}$$

Where α_{h1} and α_{h2} respectively measures the effect of migration/separation on human capital of G1 during the pre-school window and school-age window, and α_{a1} and α_{a2} measures migration/separation's effect on family attachment during these windows. Interpreting how each channel is affected by different migration/separation windows is substantially meaningful. For example, if preschool exposure matter more for emotional attachment while school-age exposure matter more for human capital, targeted policy remedies could be made to fix or prevent such effects.

To examine the importance of these two proposed mechanisms, this proposal tests whether the fertility effect of childhood parental migration/separation is fully mediated by them using the Kwon–Roth mechanism-testing framework. Specifically, define $M = (h_1, a_1)$ as the joint mediator, where h_1 and a_1 captures human capital and family attachment through measures mentioned above, respectively. The sharp null of full mediation is that, conditional on the two proposed mechanisms, childhood separation has no remaining effect on adult fertility:

$$H_0: Y_{ij}(lb, m) = Y_{ij}(lb', m) \text{ for all } lb, lb', m.$$

Where lb and lb' are possible values of the treatment, which is migration/separation here, and m is a particular value of the mediator vector. In other words, once we hold human capital and family attachment fixed, changing childhood parental migration/separation status does not change adult fertility. Rejection of this null would imply that the two measured channels do not fully explain the fertility effect, which may imply further needs for a more comprehensive model, while failure to reject would suggest that the data are consistent with full mediation through human capital and family attachment.

5.3.2. Data

Design B does not rely on any external dataset, but its identifying variation comes entirely from CHARLS families with at least two G1 children whose LB values differ. The proposal pre-specifies a three-tier downgrade rule for Design B based on the realized minimum detectable effect (MDE) at eighty-percent power and five-percent significance, benchmarked against the effect-size range documented in the adolescent-adversity and send-down literature (Yu, Yang, and Liang, 2025). Tier 1 (Design B as co-primary): MDE $\leqslant$ 0.25 child, requiring roughly five hundred discordant families with within-family LB variation and roughly one hundred same-gender discordant pairs. Tier 2 (Design B as secondary main estimate): 0.25 < MDE $\leqslant$ 0.50, allowed at the threshold of substantive meaningfulness. Tier 3 (Design B as descriptive appendix): MDE > 0.50, where the design cannot detect effects in the substantive range. The realized data place Design B on the rural × 1975-1984 candidate core in Tier 3 (79 discordant families, 40 same-gender pairs; MDE substantively above 0.50), and Design B pooled across rural cohorts in Tier 2 (408 discordant families, near but below the 500 threshold). The proposal therefore reports the within-family conditional contrast on the pooled rural sample as a secondary main estimate (Tier 2), and reports the rural × 1975-1984 within-family contrast as descriptive-appendix evidence (Tier 3).

6. Conclusion

This proposal studies whether parental migration, and consequently childhood parental separation, has long-run consequences for adult fertility in China. Using CHARLS to link the G0, G1, and G2 generations, the project develops a three-generation framework in which childhood separation affects fertility through two testable channels: human-capital formation and family attachment. Empirically, the proposal combines within-family comparisons with a set of IV strategies based on migration-cost shocks and Bartik destination-access sensitivity instruments. The IV strategies carefully distinguish the reduced-form effect of migration-induced family-arrangement changes from the isolated causal effect of separation itself, and the within-family comparison tests the different channels and windows through which the effect operates.

The expected contribution is therefore threefold: substantively, the project connects China's left-behind-child phenomenon to the country's contemporary fertility decline. Methodologically, it shows how retrospective family-history data can be used to study long-run intergenerational consequences when direct longitudinal observation is unavailable. Socially, this project sheds light on how migration policy should consider long-run fertility externalities, for instance by improving access to urban schooling, and how targeted interventions like strengthening psychological support for left-behind children are needed to repair earlier-life family disruptions that shape fertility.

Future research can extend this study in several directions. Firstly, future studies could focus on calibrating the model with merged data for richer implications, including policy counterfactuals and testing for more channels, for instance, the taste channel. Secondly, a richer structural model could explicitly incorporate the location and childcare decisions of the next generation, allowing G1 to choose whether to stay in the origin village, migrate with their children, or leave G2 in grandparental care. This would clarify how childhood separation affects not only fertility but also the living arrangements and early investments of the next generation, projecting insights into migration/separation's impact at even longer horizons.

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